

STAT 505 Assignment Submission [L3 & L4]

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1. Suppose $X = [X_1, X_2, X_3]'$ has a multivariate normal distribution with mean vector and covariance matrix

$$\mu_X = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \quad \text{and} \quad \Sigma_X = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -1 \\ 0 & -1 & 4 \end{bmatrix}$$

- (a) Find the coefficient matrix C that satisfies

$$CX = \begin{bmatrix} 4X_1 + X_3 \\ X_3 - X_2 \end{bmatrix}$$

and use it to find the mean vector and covariance matrix of CX .

1 a. $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$

$CX = \begin{bmatrix} 4X_1 + X_3 \\ X_3 - X_2 \end{bmatrix}$

$C = \begin{bmatrix} 4 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

$\mu_{CX} = C \cdot \mu_X = \begin{bmatrix} 4 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$

$4 + 2 = 6$
 $3 + 2 = 5$

$C \cdot \Sigma_X = \begin{bmatrix} 4 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -1 \\ 0 & -1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 7 & 4 \\ -2 & -6 & 5 \end{bmatrix}$

$4 + 0 + 0 = 4, 8 + 0 - 1 = 7, 0 + 0 + 4 = 4$
 $0 - 2 = -2, 0 - 5 - 1 = -6, 0 + 1 + 4 = 5$

$C' = \begin{bmatrix} 4 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix}$

$\Sigma_{CX} = C \cdot \Sigma_X \cdot C' = \begin{bmatrix} 4 & 7 & 4 \\ -2 & -6 & 5 \end{bmatrix} \cdot \begin{bmatrix} 4 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 20 & -3 \\ -3 & 11 \end{bmatrix}$

$16 + 0 + 4 = 20, 0 - 7 + 4 = -3$
 $-8 + 0 + 5 = -3, 0 + 6 + 5 = 11$

Figure 1: Handwritten work for question 1a.

The coefficient matrix C is:

$$C = \begin{bmatrix} 4 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

The mean vector of the coefficient matrix μ_{CX} is :

$$\mu_{CX} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

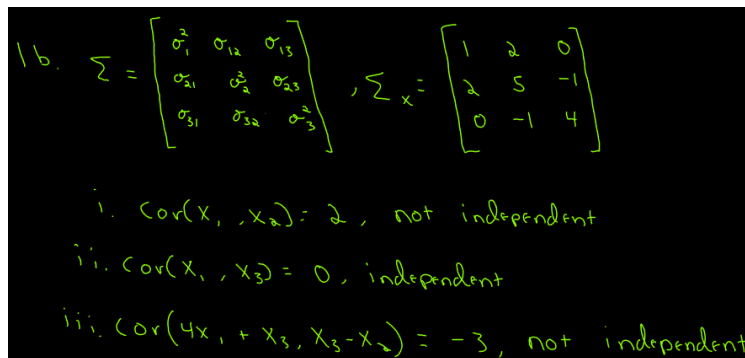
Lastly, the covariance matrix Σ_{CX} is:

$$\Sigma_{CX} = \begin{bmatrix} 20 & -3 \\ -3 & 11 \end{bmatrix}$$

(b) Determine which of the following random variables are independent. Justify your answer.

Hint: having 0 covariance is equivalent to independence for normal random variables

- i. X_1 and X_2
- ii. X_1 and X_3
- iii. $4X_1 + X_3$ and $X_3 - X_2$ *Hint: use your answer above*



1b. $\Sigma = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22}^2 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33}^2 \end{bmatrix}$, $\Sigma_x = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -1 \\ 0 & -1 & 4 \end{bmatrix}$

i. $\text{Cov}(X_1, X_2) = 2$, not independent

ii. $\text{Cov}(X_1, X_3) = 0$, independent

iii. $\text{Cov}(4X_1 + X_3, X_3 - X_2) = -3$, not independent.

Figure 2: Handwritten work for question 1b.

- i. X_1 and X_2 are dependent since the covariance is 2.
- ii. X_1 and X_3 are independent since the covariance is 0.
- iii. $4X_1 + X_3$ and $X_3 - X_2$ are dependent since the covariance is -3

2. (*adapted from J&W 1.27*) The data in “parks.csv” is of attendance (millions) at the fifteen most visited national parks and their size (acres). It can be read into SAS with the commands

```
data parks;  
infile "/parks.csv" firstobs=2 delimiter=",";  
input size visitors;  
run;
```

- (a) Construct a QQ plot relating the squared Mahalanobis distances to the chi-square quantiles. What do you expect this plot to look like if these variables were bivariate normal? Is this a reasonable assumption based on what you see here? Why or why not?

Using the following SAS Code:

```
proc princomp std out=pcresult;  
var size visitors;  
run;
```

```
data mahal;  
set pcresult;  
dist2=uss(of prin1-prin2);  
run;
```

```
proc print;  
var dist2;  
run;
```

```
proc sort;  
by dist2;  
run;
```

```
data plotdata;  
set mahal;  
prb=(_n_ - .5)/15;  
chiquant=cinv(prb,2);  
run;
```

```
proc gplot;  
plot dist2*chiquant;  
run;
```

Figure 3 depicts the QQ plot that relates the squared Mahalanobis distances to the chi-square quantiles. It is expected that the points would follow a line with a slope of one. However, the points do not follow this trend, for example with the chi-square quant being approximate 7 while the distance is 10.5.

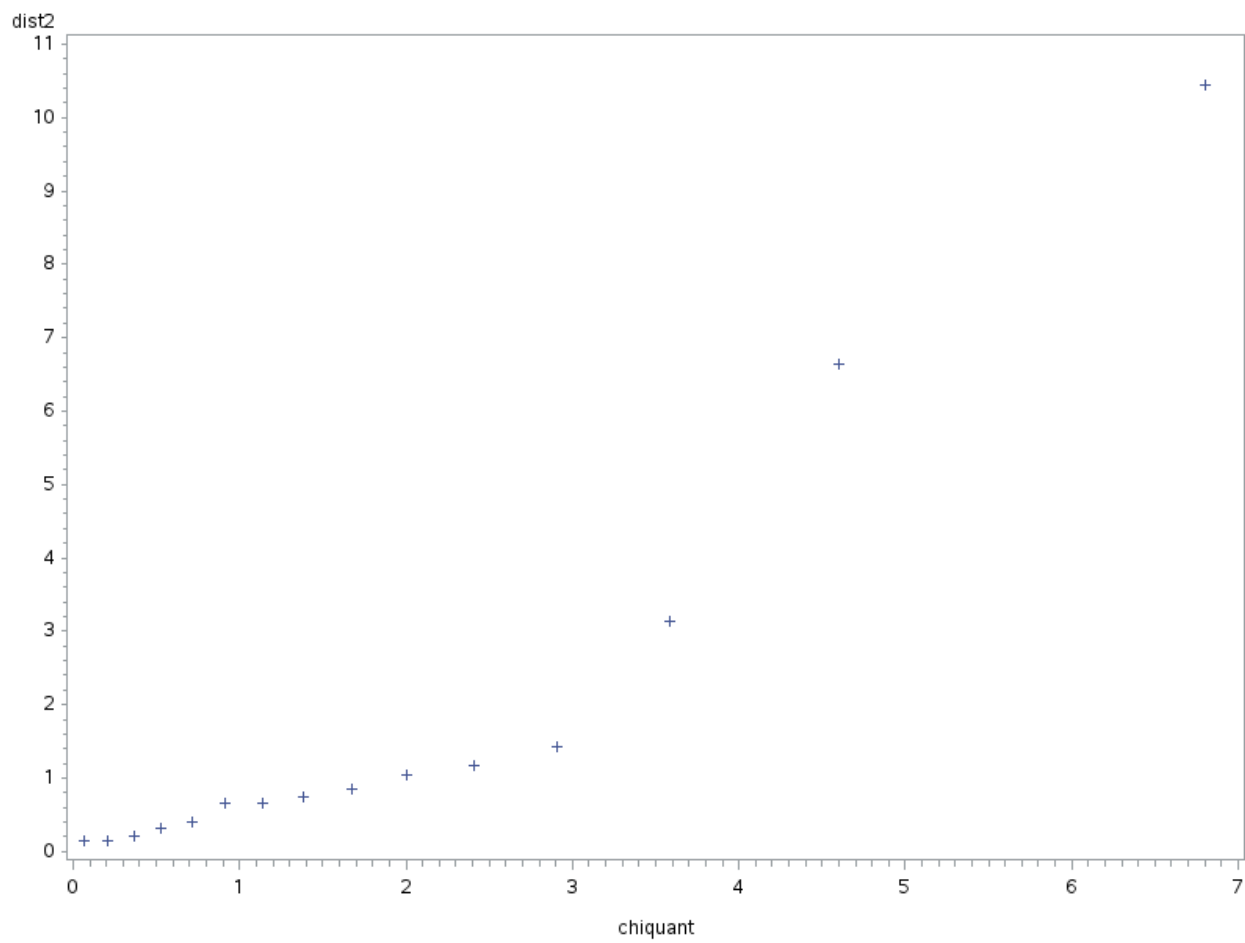


Figure 3: SAS output for question 2a.

- (b) Construct a histogram for each of the variables, and comment on the shape. Are these plots consistent with your observations above? Why or why not?

Using the following SAS Code:

```
proc univariate data=parks;  
  histogram size visitors;  
run;
```

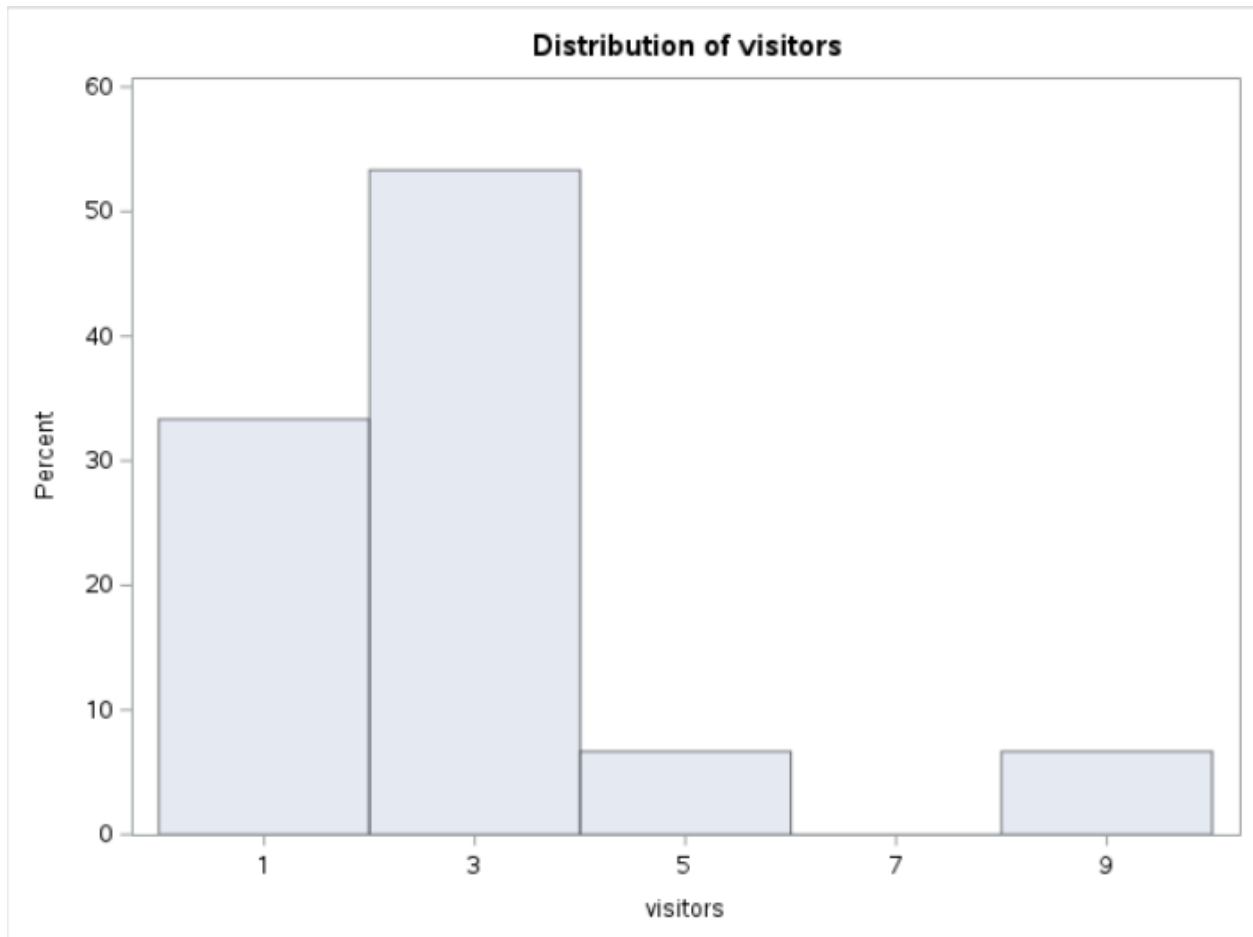


Figure 4: SAS output for question 2b(1).

Figures 4 and 5 depict histograms of the visitors and size variables. Both variables have a right skew, making it unlikely that they are normally distributed. This finding is consistent with the upward arc seen in Figure 3.

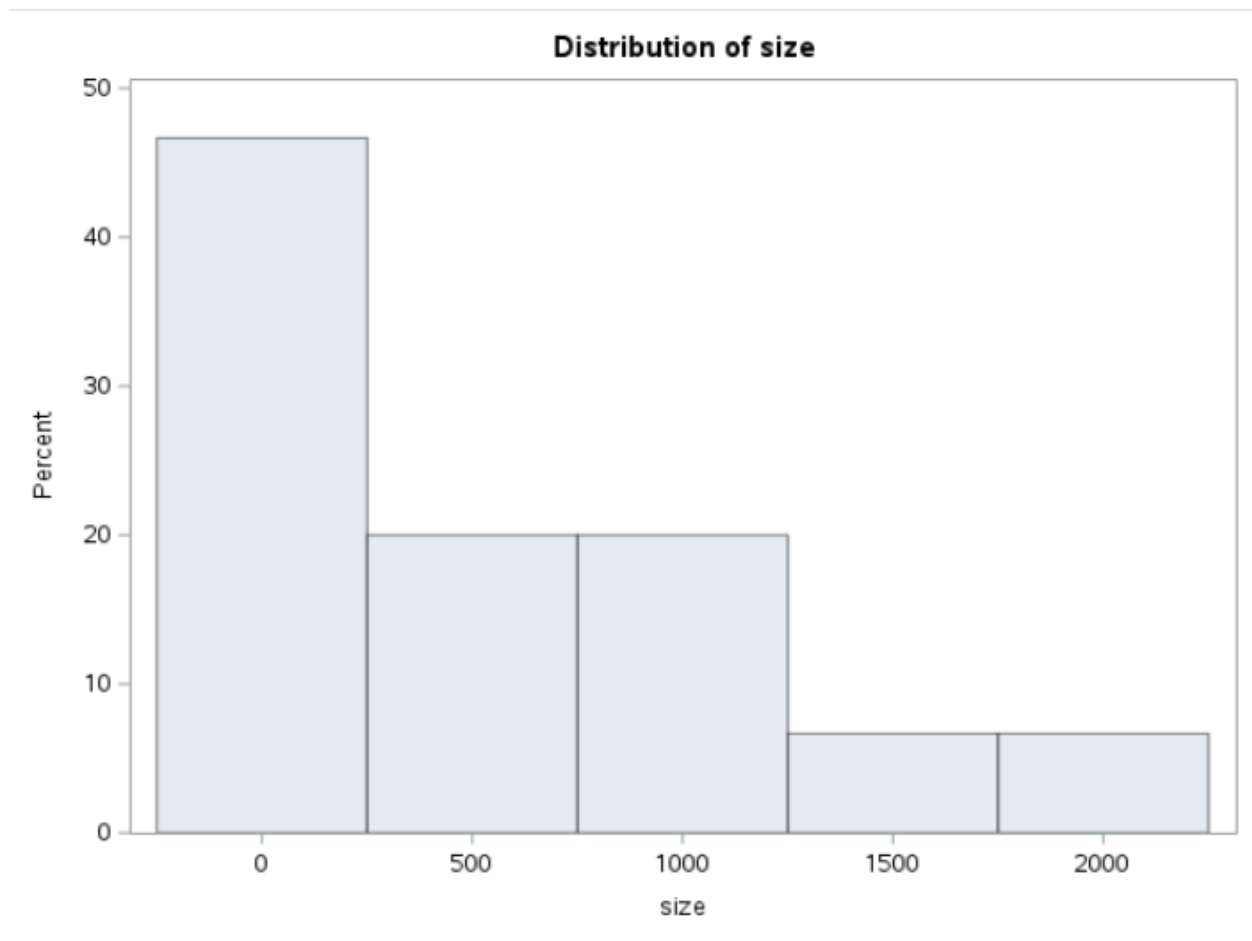


Figure 5: SAS output for question 2b(2).

- (c) Provide a scatterplot of the two variables. What shape would you expect this plot to have if these variables were bivariate normal? Is this a reasonable assumption based on what you see here? Why or why not? Also, comment on evidence of any outliers.

```
proc sgplot data=parks;  
    scatter x=size y=visitors;  
run;
```

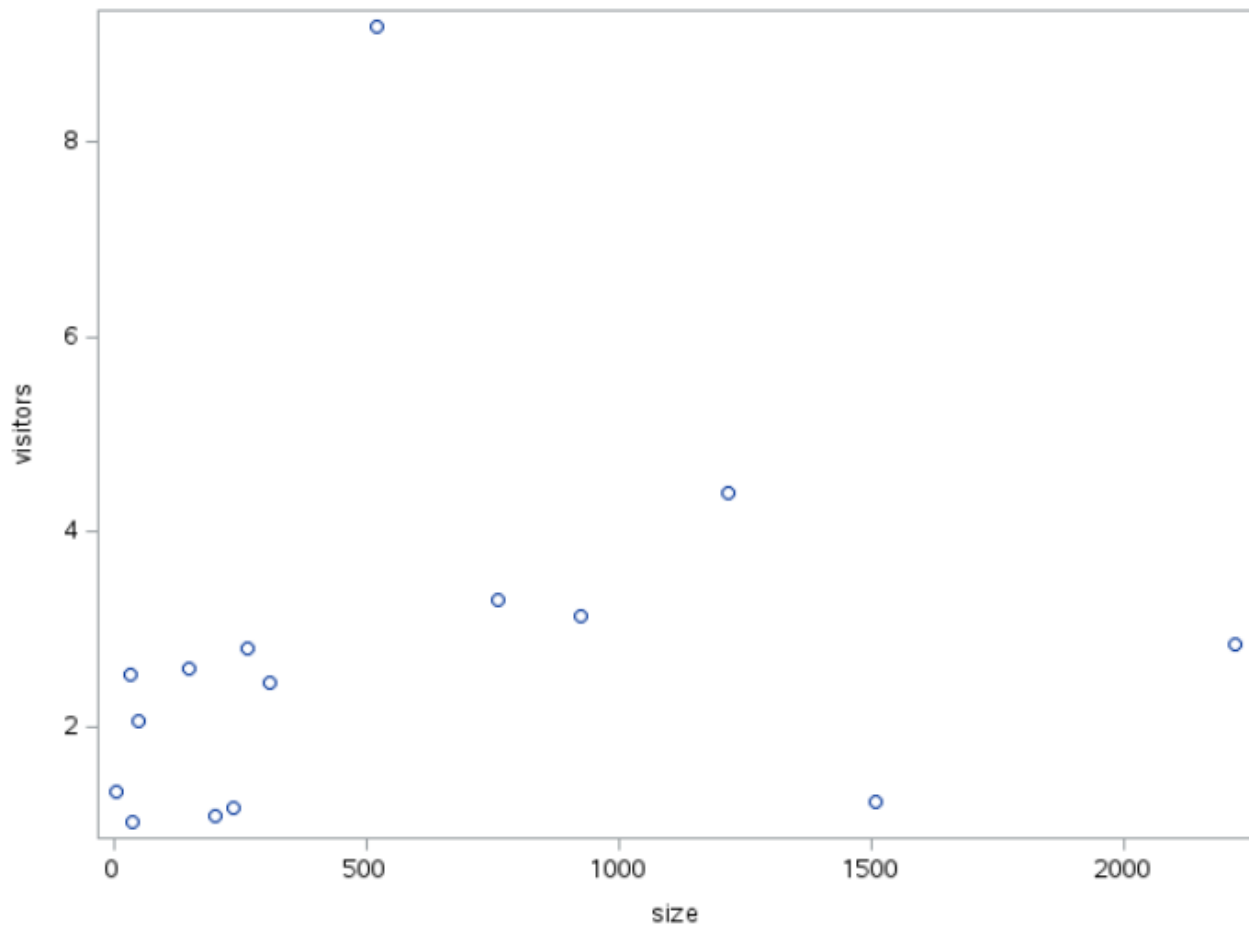


Figure 6: SAS output for question 2c.

The scatterplot in Figure 6 suggests that the two variables are not bivariate normal. It is expected that there would be an ellipse shape formed from the points in the scatter plot if there size and visitors were bivariate normal. Additionally, there are significant outliers, such as a where size exceeds 2000 and the visitors is approximately 2,000 and where the size is 500 and the visitors exceeds 8,000.